Milestone 2: Scientific Contextualization and Tradeoff Analysis

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Purpose

- This milestone places the main paper in the larger real-time systems context.
- The goal is not only to summarize related work.
- The goal is to compare assumptions, approaches, and tradeoffs.

Main Approach from the Assigned Paper

- The paper combines:
  - control performance optimization,
  - schedulability analysis,
  - rate adaptation,
  - Constant Bandwidth Server scheduling,
  - overrun handling.
- The goal is to improve control performance during normal execution.
- The system must still guarantee hard deadlines when worst-case execution occurs.

Related Approach 1: WCET-Based Scheduling

- Every task is scheduled using its WCET.
- Advantage:
  - simple to reason about,
  - safe if WCET estimates are correct,
  - strong predictability.

- Disadvantage:
  - pessimistic,
  - wastes CPU time when actual execution time is much lower than WCET,
  - can reduce sampling frequency in control systems.
- Relation to topic:
  - This is the baseline approach that overrun handling tries to improve.

Related Approach 2: Normal-Time Scheduling Without Protection

- Tasks are scheduled based on normal or average execution time.
- Advantage:
  - better average CPU utilization,
  - higher task frequency,
  - better normal-case control performance.
- Disadvantage:
  - unsafe when the task exceeds its expected time,
  - overruns can delay other tasks,
  - hard deadlines can be missed.
- Relation to topic:
  - This shows why overrun handling is needed.

Related Approach 3: Constant Bandwidth Server

- CBS assigns a budget Q_s and period T_s to a task.
- Bandwidth:
$$U_s = \frac{Q_s}{T_s}$$
- Advantage:
  - provides temporal isolation,
  - limits how much CPU time a task can consume,

- useful for variable execution time tasks.

- Disadvantage:
  - introduces scheduling overhead,
  - standard CBS may postpone deadlines too far,
  - hard-deadline guarantees require careful analysis.

- Relation to topic:
  - CBS is the main server mechanism used as the basis for overrun management.

Related Approach 4: CBShd

- CBShd is an extension of CBS for hard-deadline tasks.
- It considers the remaining computation time.
- Advantage:
  - handles partial overruns more efficiently,
  - avoids unnecessary large deadline postponements,
  - improves responsiveness.
- Disadvantage:
  - requires estimating remaining computation time,
  - still introduces runtime overhead,
  - implementation is more complex than simple WCET scheduling.

Related Approach 5: Resource Sharing Protocols

- Real tasks may share resources such as:
  - memory buffers,
  - sensors,
  - communication buses,
  - hardware peripherals.
- Shared resources can cause blocking.
- The Stack Resource Policy can be used to bound blocking times.
- Tradeoff:
  - resource protocols improve predictability,
  - but increase analysis complexity.

Tradeoff: Safety vs Efficiency

- WCET-only scheduling:
  - high safety,
  - low efficiency if WCET is rare.
- Normal-time scheduling:
  - high efficiency,
  - unsafe without overrun protection.
- CBS/CBShd:
  - tries to balance both,
  - improves efficiency while preserving bounded interference.

Tradeoff: Precision vs Runtime Overhead

- Small server budget:
  - more precise overrun handling,
  - more frequent deadline updates,
  - higher overhead.
- Large server budget:
  - lower overhead,
  - less precise handling of small overruns.

Tradeoff: Offline Guarantees vs Online Adaptation

- Offline analysis:
  - checks schedulability before runtime,
  - provides stronger guarantees.
- Online adaptation:
  - reacts to actual runtime behavior,
  - improves flexibility,
  - harder to guarantee.
- The assigned paper combines both:
  - offline guarantees using WCET,
  - online overrun handling through server-based scheduling.

Embedded Systems Consequences

- The approach is relevant for embedded systems with variable workloads.
- Good fit:
  - robotics,
  - sensor processing,
  - visual tracking,
  - radar tracking,
  - autonomous systems.
- Challenges:
  - WCET measurement,
  - scheduler overhead,
  - limited CPU resources,
  - resource sharing,
  - implementation on Linux or microcontrollers.

Selected References and Why They Are Relevant

- Caccamo, Buttazzo, Sha:
  - main paper,
  - rate adaptation,
  - overrun handling,
  - CBShd.
- Buttazzo book:
  - official background,
  - real-time scheduling,
  - overload handling,
  - resource reservations.
- Abeni and Buttazzo:
  - original CBS reference.
- Liu and Layland:
  - foundational EDF and real-time scheduling theory.

- Baker:
  - Stack Resource Policy and resource sharing.